



Invisible Ink Letter

Secret Messages with Lemon Juice!



Lemon Juice

Our invisible ink ingredient!



30 Minutes

Hands-On Science Session



Heat Reveals

Oxidation turns ink brown!

Write a secret message with lemon juice — watch it disappear! Then use heat to reveal it like magic. This is real science: INVISIBLE INK, ACID, and OXIDATION all in one amazing experiment!

That looks like MAGIC — but it is actually SCIENCE! ■



■ Warm-Up: Can You See This?

Can you see the water writing on the floor? Sometimes, things are **INVISIBLE** — we cannot see them! Lemon juice dries just like water writing on a floor — it disappears! But we have a science secret to make it come back...

■ **Water writing dries on a hot floor** — it becomes INVISIBLE!

■ **Wind** — you cannot see it — but you can feel it!

■■ **Heat from the sun dries wet clothes** — heat has power!

■ Three Magic Science Words!

■ INVISIBLE

When something is invisible, we CANNOT see it. The lemon juice dries clear on paper.

Example: Water writing on a hot stone floor — it dries and disappears!

■ ACID

Acids are sour liquids with special powers! Lemon juice is a mild acid called CITRIC ACID.

Example: Nimbu in dal, Imli in sambar — sour foods contain natural acids!

■ OXIDATION

When heat or air changes things and they turn BROWN. Heat causes a colour change!

Example: When you cut an apple and leave it out, it turns brown - that is OXIDATION every day!



■ Prediction Time!

■ My Prediction:

I think when we heat the paper, the lemon juice will _____

■ Every guess is a good guess! Scientists ALWAYS predict before they try. Return to your prediction after the experiment to see if you were right!

■ Materials for Our Secret Science Experiment



Fresh lemons (2–3)

Squeeze juice before session. Cost: approx. ■10–■15



Cotton buds / thin paintbrush

1 per child. Available at any medical/general store.



White A4 paper

2 per child — one practice, one final secret message.



Hair dryer (teacher use only)

OR warm light bulb. NEVER open flame!



Secret Scientist badge

1 per child as end-of-session reward.

■ BUDGET NOTE: Total material cost per student group approx. ■10. Indian alternative to lemon: onion juice or vinegar (also safe acids).



Writing Our Secret Message

- 1 Dip your cotton bud into the lemon juice. Just a little — do not overload!
- 2 Write a secret message or draw a simple picture on your WHITE paper.
- 3 Let your message DRY completely. Hair dryer on LOW (teacher only) speeds this up.
- 4 Check: Can you still see your writing? No? PERFECT — invisible ink is ready!

Drying Time — Invisible Ink is Ready!

Now our secret message is INVISIBLE! We cannot see it yet. Can you still see your message? No? That is invisible ink science! The lemon juice dried clear — just like water writing on a stone floor.

The Big Reveal! Apply Heat!

HEAT SOURCE	HOW TO USE	SAFETY LEVEL
Warm 60W light bulb	Hold paper 15–20 cm above	■ SAFEST
Clothes iron (LOW)	Teacher slides across paper	■ Safe (teacher only)
Hair dryer (HIGH)	Hold 10 cm from paper	■ Safe (teacher only)

3... 2... 1... WOW! The secret message appears! Invisible ink turns brown with heat because of OXIDATION! The lemon juice acid reacts with heat — just like roti turning golden-brown on the tawa! That is NOT magic — it is SCIENCE!



■ What is Invisible Ink?

Lemon juice is a clear liquid called an **ACID**. When it dries, it is **INVISIBLE** — we cannot see it. **HEAT** makes the lemon juice turn brown and show the secret! Like writing with water on a dry stone floor — it disappears when dry, but heat brings the lemon juice back as brown writing!

■ What is Oxidation?



Banana turning brown



Avocado going brown after cutting



Butter smelling bad

■ What is Acid?

Acids are sour liquids with special powers! Lemon juice is an acid called **CITRIC ACID**. Acids are like secret agents — they hide but have special powers that come out when conditions are right!

■ Nimbu in Dal

■ Imli in Sambar

■ Raw Mango
(Kaccha Aam)



■ ■ Safety First!

- **NEVER** allow children to hold paper over an open flame or candle.
- **ONLY** the teacher applies heat — hair dryer, iron, or warm bulb.
- **Do NOT** eat the lemon juice or put it in eyes — rinse with water if contact.
- In summer, sunlight for 5–8 minutes works as a safe heat alternative.

■ Session Outline — 30 Minutes

WARM-UP 4 MINS

Can You See This? Activate curiosity with invisible things.

CONCEPT 5 MINS

Learn INVISIBLE, ACID and OXIDATION with Indian examples.

PREDICTION 2 MINS

Make a smart guess before the experiment begins!

EXPERIMENT 7 MINS

Write secret message with lemon juice. Let it dry invisible!

THE REVEAL 6 MINS

Apply heat — watch the message appear! 3... 2... 1... WOW!

WRAP-UP 6 MINS

Reflection, vocabulary recap, and Secret Scientist badge!

■ Wrap-Up and Reflection

INVISIBLE

What is invisible ink? Tell me in one sentence!

ACID

What is an acid? Name one example from home!

OXIDATION

What happened to the lemon juice when it got hot?



■ Success and Secret Scientist Badge!

- Every student made at least ONE prediction before the experiment.
- At least 80% of students can say the word INVISIBLE in a sentence.
- Students watched the reveal quietly and reacted with genuine surprise.
- At least one student connects colour change to a real-life example.
- Every student completes the drawing on the activity sheet.
- Students used the word ACID at least once.
- NO child touched the heat source at any point during the session.

■ What's Next in Our Science Adventure?

SESSION 2

What Else Makes Invisible Ink?: Orange juice, vinegar, milk — test them all!

SESSION 3

Why Do Things Turn Brown?: Oxidation in fruits — apple, banana, mango!

SESSION 4

Acids and Bases: Red cabbage changes colour with lemon vs soap!

SESSION 5

Baking Soda Volcano: Two substances react — gas is produced!

CAPSTONE

Science Fair Corner: Students reveal their own secret message to parents!





**Thank you for exploring invisible ink with us! You
are all Secret Scientists now! ■■■**

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